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AMIRTHAA DAIRY PVT. LTD.
Erode-638104, Tamil Nadu, India
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Date: 21.10.2026

CERTIFICATE OF MILK SAMPLE SUPPLY, RESEARCH PERMISSION, AND EXPERIMENTAL PROTOCOL ACKNOWLEDGEMENT

To Whomsoever It May Concern

This is to certify that **Amirthaa Dairy Pvt. Ltd., Erode, Tamil Nadu, India**, has permitted Mr. Anbumani V, Assistant Professor (Selection Grade), Department of Electronics and Communication Engineering, Kongu Engineering College, Perundurai, Tamil Nadu, India, to obtain and use cow and buffalo milk samples for the research study entitled **"AI-Driven RGB-Temporal Image Processing Framework for Real-Time Milk Quality Assessment Using Edge Computing."**

The milk samples were obtained for research purposes involving the Methylene Blue Reduction Test (MBRT), temperature-controlled incubation, image acquisition, RGB-temporal image processing, and machine-learning-based classification of milk-quality categories. The experimental work involves monitoring colour changes in milk samples under controlled laboratory conditions and classifying the observed MBRT responses as Fresh, Mildly Spoiled, or Spoiled.

We hereby confirm the following:

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1. **Permission for sample collection and use:** Amirthaa Dairy Pvt. Ltd. has granted permission for the collection or supply of cow and buffalo milk samples for the above-mentioned research study.
2. **Research purpose:** The samples may be used for MBRT-based experimental analysis, image acquisition, colour-feature extraction, machine-learning model development, and scientific publication of the research findings.
3. **Experimental procedure acknowledgement:** The proposed experimental procedures involving the collected milk samples (**Cow, and buffalo**) including MBRT reagent treatment, incubation at approximately (37°C), periodic image acquisition, and subsequent image analysis, have been explained to and acknowledged by the authorized representative of the company.
4. **Animal handling:** The research activities conducted by the research team using the supplied milk samples are laboratory-based. **No animals were subjected to additional experimental procedures by the research team as part of this study.**
5. **Publication permission:** The company has no objection to the use of the samples and the resulting research data for academic and scientific publication, provided that confidential business information is not disclosed without authorization.

This certificate is issued at the request of the researcher for documentation of sample provenance, permission for research use, and submission of the associated research manuscript to *Scientific Reports*, Springer Nature.

The above statements reflect the permission granted by the company and the information available to its authorized representative.

Note: Neither the research team nor Amirthaa Dairy Pvt. Ltd., Erode, performed any additional experimental procedures on live animals; all experimental procedures were conducted exclusively on the collected milk samples.

For Amirthaa Dairy Pvt. Ltd.
S.P. Mani & Mohan Dairy (India) Pvt. Ltd.,
1785/1, Kulur Panchayat, Kathakinaru Village.
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